

Warm-up

- (1) Write the Cartesian point $(\sqrt{3}, -1)$ in polar coordinates.
- (2) Write the polar point $(6, 5\pi/3)$ in Cartesian coordinates.

Medium

- (1) Find a (cartesian, non-parametric) equation for a line tangent to the polar curve given by $r = 2 + \cos(\theta)$ at the polar coordinates $(5/2, \pi/3)$.
- (2) Find the area inside the cardioid $r = 1 + \cos(\theta)$.
- (3) Find the length of the polar curve $r = e^{3\theta}$ from $\theta = 0$ to $\theta = 2$.
- (4) Find the area between a large loop and a small loop of the curve $r = 1 + 2\cos(3\theta)$.

Stretch

- (1) The polar curve $r = 15/(8\sin(\theta) + 17\cos(\theta))$ is actually a straight line. Write it in Cartesian coordinates as $y = mx + b$.
- (2) Generalize (1) above—find the polar equation for a line $y = mx + b$ with $b \neq 0$.